

1 The Binary System in the Plane of the Orbit

Consider a binary system, comprised of stars 1 and 2, with masses M_1 and M_2 . Each star orbits about the center-of-mass (CM) of the system along an elliptical trajectory. If we let $\mathbf{r}_1$ and $\mathbf{r}_2$ be the position vectors of stars 1 and 2 relative to the CM, then it can be shown that the relative position vector, $\mathbf{r} \equiv \mathbf{r}_1 - \mathbf{r}_2$, satisfies

$$r \equiv |\mathbf{r}| = \frac{a(1 - e^2)}{1 + e \cos \theta} , \quad (1)$$

where a and e are, respectively, the semimajor axis and eccentricity of the relative orbit. The angle θ is called the *true anomaly* and is measured in the orbital plane, from the direction of periastron, in the sense defined by the angular momentum vector of the binary (using the right-hand rule). The trajectory of either star about the CM is obtained using the relation $M_1\mathbf{r}_1 - M_2\mathbf{r}_2 = 0$, from which it follows that

$$\mathbf{r}_1 = \frac{M_2}{M} \mathbf{r} ; \mathbf{r}_2 = -\frac{M_1}{M} \mathbf{r} , \quad (2)$$

where $M = M_1 + M_2$. It is clear from eq. (2) that the trajectories of stars 1 and 2 are ellipses with eccentricity e and semimajor axes $a_1 = aM_2/M$ and $a_2 = aM_1/M$, respectively.

2 The Binary System in the Plane of the Sky

In practice, we only see the binary system projected onto the plane of the sky. Two angles are required to describe the orientation of the true elliptical orbit of either star relative to the direction of the line of sight (LOS) and the plane of the sky. The *inclination angle*, $0^\circ < i < 90^\circ$, is the angle between the orbital plane and the plane of the sky, so that $i = 0^\circ$ means that we are seeing the binary orbit face-on, and $i = 90^\circ$ means that we are seeing the orbit edge-on. The second angle, which I will call the *angle of periastron*, $0^\circ < \varphi < 360^\circ$, is the angle between the direction of periastron and the line of nodes (LON; where the orbital plane intersects the plane of the sky). The angle φ is measured from a particular direction along the line of nodes to the direction of periastron, in the sense defined by the projection of the angular momentum vector onto the LOS. For instance, suppose that the LON is along the north-south direction on the sky, in which case φ could be measured from the northward direction to the direction of periastron, in the sense defined by the projected orbital motion (see Fig. ??).

3 The Line-of-Sight Velocity of a Binary Component

Most binaries are too far away to be resolved into individual stars. Therefore, it is most often the case that the binary motion is manifested by eclipses or as Doppler shifting of either spectral lines, in the case of a star, or the frequency of a periodic pulse, as in the case of a binary pulsar. Only the Doppler shifting of spectral lines or pulses will be discussed here.

Figure 1: The LOS velocity for $i = 90^\circ$.

Suppose, for the sake of argument, that we are observing star 1 of a binary system. For LOS velocities of star 1, $v_{1,\text{los}}$, much less than the speed of light, c , the Doppler shift of the frequency is just

$$-\frac{\Delta\nu}{\nu_0} \simeq \frac{v_{1,\text{los}}}{c} , \quad (3)$$

where ν_0 is the rest frequency of the radiation or periodic pulse.

For a Keplerian orbit, with known orientation angles, i and φ , it is straightforward to compute $v_{1,\text{los}}$. First, consider the case where the orbit is seen edge-on ($i = 90^\circ$; Figure ?? illustrates the geometry). From Fig. ?? it is evident that the LOS projection of the distance of star 1 from the CM is $r_1 \sin(\varphi + \theta)$, from which follows the LOS projection of the velocity of star 1,

$$v_{1,\text{los}} = \dot{r}_1 \sin(\varphi + \theta) + r_1 \dot{\theta} \cos(\varphi + \theta) . \quad (4)$$

Differentiating eq. (1) gives

$$\dot{r}_1 = \frac{a_1(1 - e^2)}{(1 + e \cos \theta)^2} e \sin \theta \dot{\theta} . \quad (5)$$

Recall that for a Keplerian orbit, $r^2 \dot{\theta}$ is the constant angular momentum per unit reduced mass of the system:

$$r^2 \dot{\theta} = [GMa(1 - e^2)]^{1/2} \equiv \Omega a^2 (1 - e^2)^{1/2} , \quad (6)$$

where $\Omega = (GM/a^3)^{1/2}$ is the Keplerian orbital frequency. It follows that $\dot{r}_1$ is simply

$$\dot{r}_1 = \frac{\Omega a_1}{(1 - e^2)^{1/2}} e \sin \theta . \quad (7)$$

Substituting for $\dot{r}_1$ and $\dot{\theta}$ in eq. (??) gives

$$\begin{aligned} v_{1,\text{los}} &= \frac{\Omega a_1}{(1-e^2)^{1/2}} [e \sin \theta \sin(\varphi + \theta) + (1 + e \cos \theta) \cos(\varphi + \theta)] \\ &= \frac{\Omega a_1}{(1-e^2)^{1/2}} [e(\cos \theta \cos(\varphi + \theta) + \sin \theta \sin(\varphi + \theta) + \cos(\varphi + \theta))] . \end{aligned} \quad (8)$$

Using the identity $\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$, we obtain the result

$$v_{1,\text{los}} = \frac{\Omega a_1}{(1-e^2)^{1/2}} [e \cos \varphi + \cos(\varphi + \theta)] . \quad (9)$$

It is a trivial matter to introduce an inclination $i < 90^\circ$ into eq. (??): simply multiply by $\sin i$. A vector pointing along the LOS for $i = 90^\circ$ will make an angle $90^\circ - i$ with the LOS for $i < 90^\circ$, so that we must multiply the original vector by $\cos(90^\circ - i) = \sin i$. We now have the final result for the LOS projection of the orbital velocity of star 1:

$$v_{1,\text{los}} = \frac{\Omega a_1 \sin i}{(1-e^2)^{1/2}} [e \cos \varphi + \cos(\varphi + \theta)] . \quad (10)$$

4 Parametric Solution of the Kepler Problem

Note that eq. (1) gives the relative separation r as a function of the polar angle θ , but makes no reference to the elapsed time, which might, for instance, be measured relative to the time of periastron passage. It turns out that there is no closed-form expression for $r(t)$; however, there is a convenient parametric solution to the Kepler problem. The parametric solution can be obtained by a suitable change of variables when integrating

$$\int dt = \int dr \dot{r}^{-1} = \pm \int dr \left[2 \left(\mathcal{E} + \frac{GM}{r} \right) - \frac{h^2}{r^2} \right]^{-1/2} , \quad (11)$$

where $\mathcal{E}$ and h are the energy and angular momentum per unit reduced mass, respectively. However, we will adopt a more geometric approach.

All of what follows relies heavily on Figure ??, so take a careful look. The interior ellipse represents the relative orbit of the two stars, the angle θ is the true anomaly defined in §1, and the angle η is known as the *eccentric anomaly*. The circle that encloses the ellipse has radius a , and η is measured from the direction of periastron, in the sense of the orbital motion, just like θ .

There is an important and useful mathematical relationship between the ellipse and the circle that encloses it. Let (x_c, y_c) be a point on the circle. The coordinates x_c and y_c satisfy the usual equation for a circle,

$$\frac{x_c^2}{a^2} + \frac{y_c^2}{a^2} = 1 . \quad (12)$$

Now, for a given x -coordinate, $x = x_c$, what is the corresponding y -coordinate, y_e , on the ellipse (in the same half-plane as y_c)? With $x = x_c$, y_e is obtained from the equation for an ellipse,

$$\frac{x_c^2}{a^2} + \frac{y_e^2}{b^2} = 1 , \quad (13)$$

Figure 2: The geometry behind the eccentric anomaly.

where $b = a(1 - e^2)^{1/2}$ is the semiminor axis of the ellipse. If we compare eqs. (??) and (??), we immediately see that $y_e = y_c(1 - e^2)^{-1/2}$, so that y_e and y_c are just related by the scale factor $(1 - e^2)^{-1/2}$. This result is not too surprising if we recall that an ellipse is just a projected circle, with a semimajor axis equal to the circle's radius.

An inspection of Fig. ?? reveals that we can write r in terms of η :

$$\begin{aligned} r^2 &= \overline{OB}^2 + \overline{BC}^2 \\ &= (a \cos \eta - ae)^2 + ((1 - e^2)^{1/2} a \sin \eta)^2 \\ &= a^2(\sin^2 \eta + \cos^2 \eta + e^2 - e^2 \sin^2 \eta - 2ae \cos \eta) \\ &= a^2(1 + e^2 \cos^2 \eta - 2ae \cos \eta) \\ &= a^2(1 - e \cos \eta)^2, \end{aligned}$$

where $\overline{BC} = (1 - e^2)^{1/2} \overline{BD}$ has been used to obtain the second line. Therefore, we have

$$r = a(1 - e \cos \eta). \quad (14)$$

Equations (1) and (??) give the relation between θ and η . It is left as an exercise to the reader to show that

$$\tan \frac{\theta}{2} = \left(\frac{1 + e}{1 - e} \right)^{1/2} \tan \frac{\eta}{2}. \quad (15)$$

(Hint: use the identity $\tan^2 \alpha/2 = (1 - \cos \alpha)/(1 + \cos \alpha)$.)

The scale relation between y_e and y_c for a given x -coordinate applies to areas as well. For instance, the area within the ellipse swept out by the angle θ , which I will denote by $\widehat{AOC}$, is just $(1 - e^2)^{-1/2}$ times the area within the circle, $\widehat{AOD}$, swept out by the angle $\angle AOD$. (If this is not obvious to you, imagine computing the area under the curve (the ellipse or the circle) by summing the areas of tall, thin rectangles.) The area $\widehat{OAD}$ is computed in two

parts:

$$\begin{aligned}
\widehat{AOD} &= \underbrace{\frac{1}{2}a^2\eta}_{\text{sector swept by } \angle AED} - \underbrace{\frac{1}{2}ae \times a \sin \eta}_{\text{triangle } OED} \\
&= \frac{a^2}{2}(\eta - e \sin \eta) ,
\end{aligned} \tag{16}$$

so that

$$\widehat{AOC} = \frac{a^2}{2}(1 - e^2)^{1/2}(\eta - e \sin \eta) . \tag{17}$$

Now, think back to a time when things were simpler, and Kepler's three laws of planetary motion were the most complicated physics you had to deal with. You may recall Kepler's second law, which states that equal areas are swept out in equal times by the position vector that points from the CM of the system to the star in question. The relation between area, Σ , and time, t , is quite simple:

$$\dot{\Sigma} = \frac{\pi ab}{P} = \frac{\pi a^2(1 - e^2)^{1/2}}{P} , \tag{18}$$

where πab is the area of the ellipse, and P is the orbital period. If we choose to measure time relative to the time of periastron passage, t_p , then inspection of eqs. (??) and (??) reveals that

$$t - t_p = \frac{P}{2\pi}(\eta - e \sin \eta) . \tag{19}$$

Oftentimes, you will see various quantities related to the binary motion plotted not as a function of time, but rather as a function of the *mean anomaly*, $M = 2\pi(t - t_p)/P < 2\pi$, sometimes referred to as the *orbital phase*. As far as I know, the mean anomaly has no simple geometric interpretation.